

Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 35

Title: Lysosomal Storage Disorders

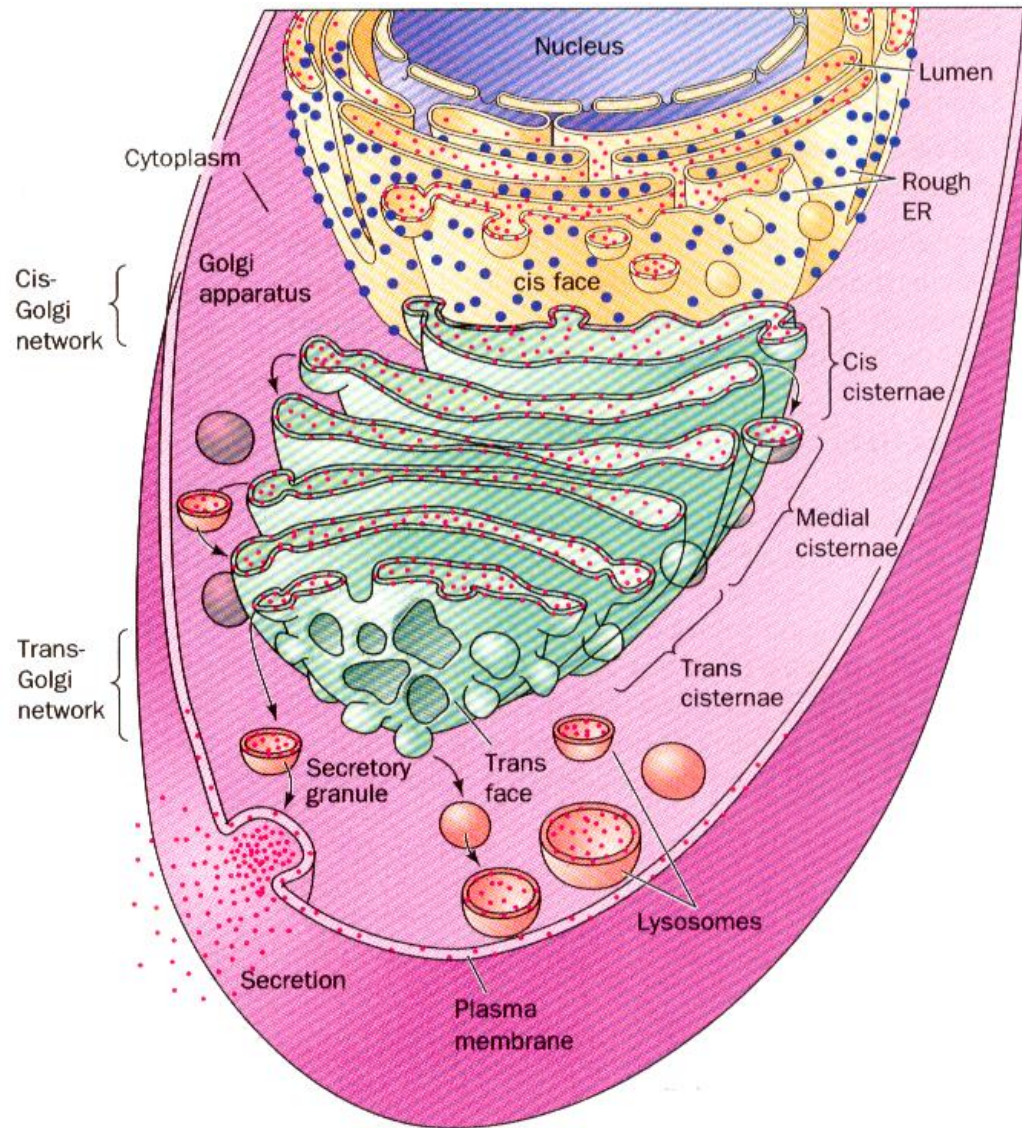
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Objectives: Lysosomal Storage Disorders

SOM. MK.I.BPM2.3.DM.3.BCHM.1334	Describe metabolic basis of hepatosplenomegaly, intellectual disability, macular cherry red spot in patients with lysosomal storage disorders.
SOM. MK.I.BPM2.3.DM.3.BCHM.1335	Distinguish between lysosomal storage diseases based on deficient enzyme, accumulating compound, clinical features, general cell appearance and clinical features: Hunter vs Hurler syndrome; Tay-Sachs disease; Fabry disease; Gaucher disease; Niemann-Pick disease; Metachromatic leukodystrophy; I-cell disease

Recommended Reading

- Lippincott's Biochemistry 9th Edition
 - <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Lippincott Illustrated Reviews: Biochemistry, 9th Edition
 - p. 179-180 (Chapter 14, sections IV. Degradation & V. Mucopolysaccharidoses and figure 14.12 and questions 14.1-14.5),
 - [Glycosaminoglycans, Proteoglycans, and Glycoproteins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - P. 233-6 (Chapter 17, section VII. D Sphingolipidosis and figures 17.17, 17.18, 17.19 & 17.20 and question 17.3)
 - [Phospholipid, Glycosphingolipid, and Eicosanoid Metabolism | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - Lysosomal Storage Diseases - Goldman-Cecil Medicine
 - [Lysosomal Storage Diseases - Goldman-Cecil Medicine - ClinicalKey Student](#)

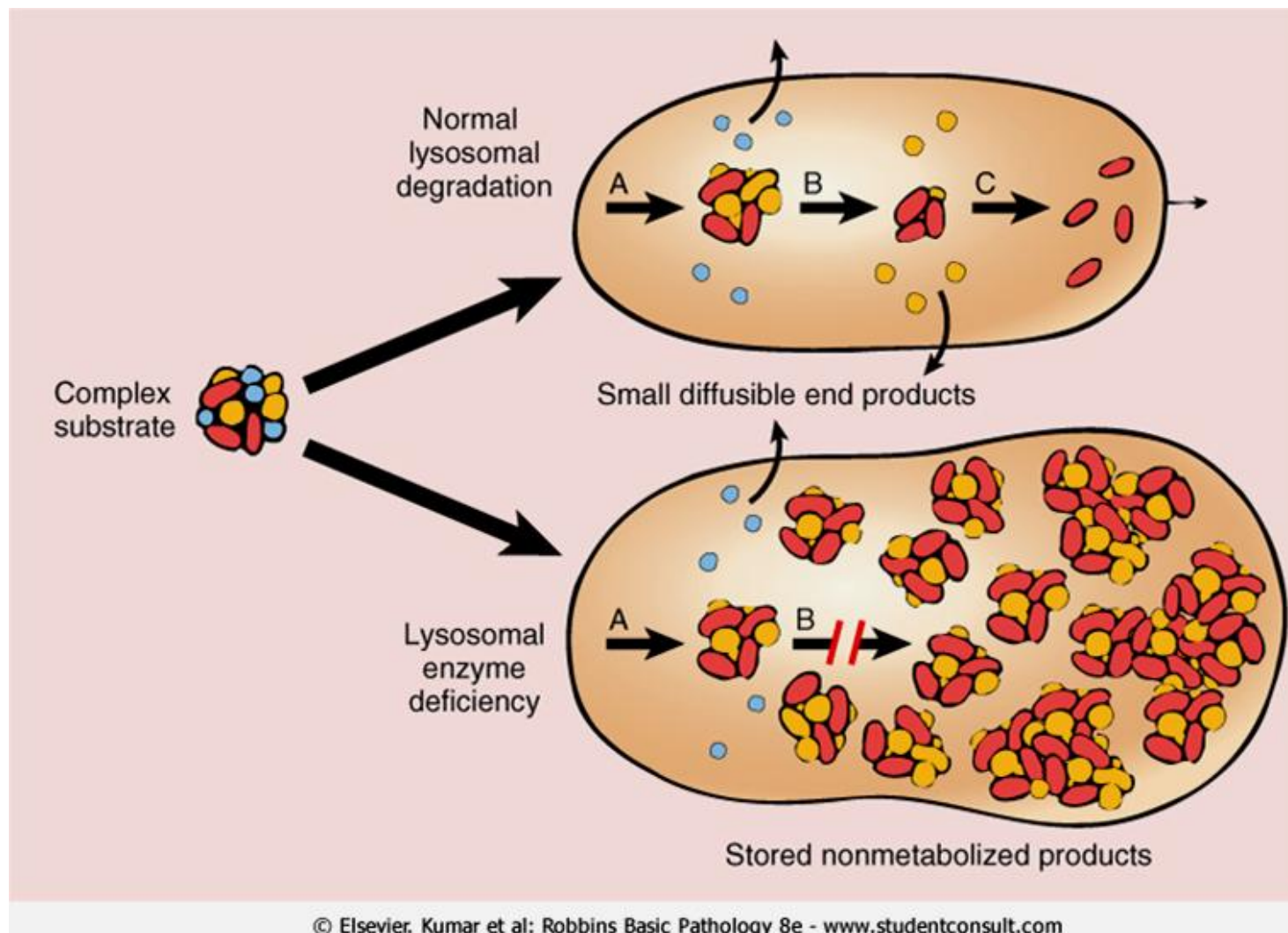


Lysosomes are subcellular organelles involved in the degradation of **complex lipids, glycosaminoglycans, and glycogen.**

Generally, the deficiency of a specific lysosomal enzyme involved in the catabolism of the compounds results in lysosomal storage disorders (LSD)

Lysosomal enzymes

- Lysosomal enzymes are active at an acidic pH (pH 5)
- They are involved in the degradation of **glycosaminoglycans** (mucopolysaccharides), **sphingolipids** (glycolipids and sphingophospholipids) and **glycogen**.
- These enzymes remove one sugar unit at a time from these molecules
- A specific enzyme of the degradative pathway is deficient in these disorders, resulting in **accumulation of the substrate** of the pathway
- The rate of biosynthesis of the compound is usually normal, but the **rate of lysosomal degradation is slow** due to inherited deficiency of the lysosomal enzyme
- The substrate of the deficient enzyme accumulates in the lysosomes, resulting in swelling of the lysosomes and consequent **organ damage**



Biochemical basis of lysosomal storage disorders

A complex substrate is normally degraded by a series of lysosomal enzymes (**A**, **B**, and **C**) into soluble end products. If there is a deficiency or malfunction of one of the enzymes (e.g., **B**), catabolism is incomplete, and insoluble intermediates accumulate in the lysosomes.

Common features of lysosomal storage disorders

- Most of the disorders are **autosomal recessive disorders**
- Commonly affect infants and **young children**
- Storage of insoluble intermediates in the mononuclear phagocyte system, giving rise to **hepatosplenomegaly**
- Frequent **CNS involvement** with associated neuronal damage
- Cellular dysfunctions, caused by
 - Storage of undigested material
 - And as a result of macrophage activation and release of cytokines
- Confirmation of diagnosis by assay of the specific enzyme in **leukocytes or skin fibroblasts**

Lysosomal storage disorders

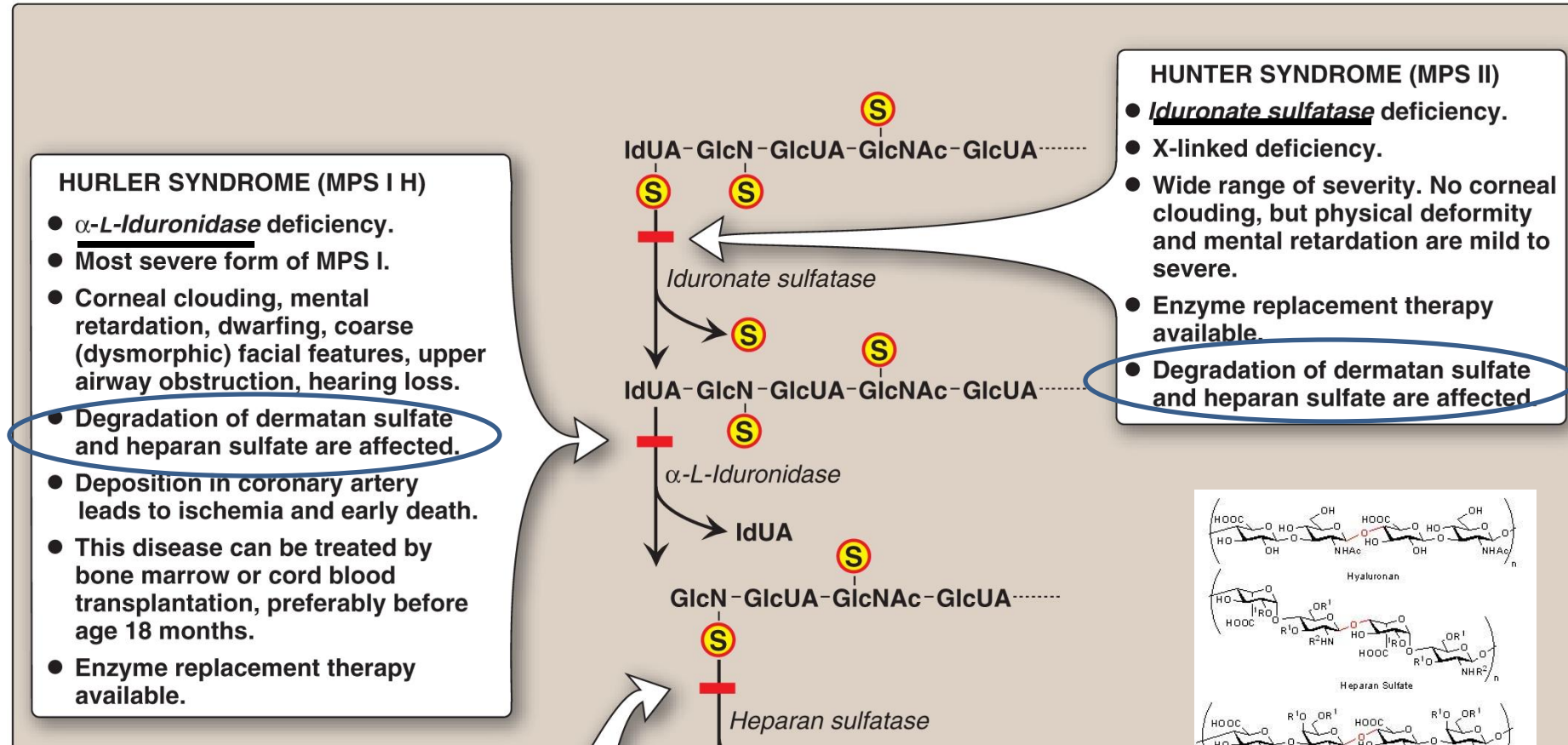
- **Mucopolysaccharidoses**
 - Hurler syndrome 🦶
 - Hunter syndrome 🦶
- **Sphingolipidoses**
 - Tay-Sachs disease
 - Gauchers disease 🦶
 - Fabry disease 🦶
 - Niemann-Pick disease (sphingophospholipid) 🦶
 - Metachromatic leukodystrophy
- **Glycogen storage disease (Pompe's disease, GSD II) 🦶**
- **I-cell disease**

For each of the disorder listed above – Identify the compound accumulated and the biochemical defect (enzyme), clinical features, cell morphology

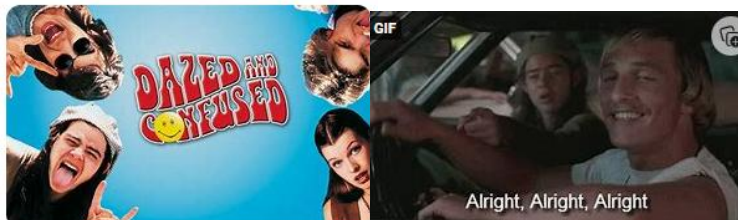
Mucopolysaccharidoses

- Characterized by a deficiency of an enzyme required for the lysosomal degradation of **glycosaminoglycans** (mucopolysaccharides)
- The glycosaminoglycans are degraded in the lysosomes by the removal of one sugar unit at a time
- **Hurler syndrome** and **Hunter syndromes** are characterized by accumulation of **dermatan sulfate and heparan sulfate** (glycosaminoglycans)
- Enzyme replacement therapy with the specific enzyme has been tried in both the disorders with varied success (Does not cross BBB)

Heparan sulfate and dermatan sulfate degradation



“I hurl a rock at your head and put you in a daze”

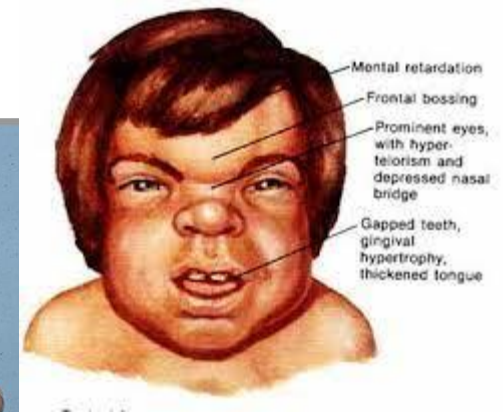
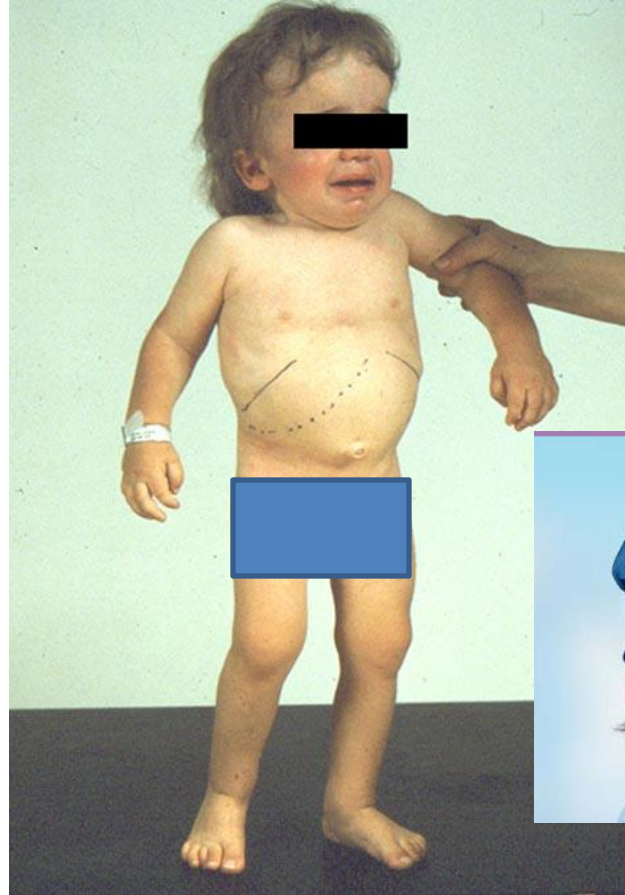


Lippincott's p.164

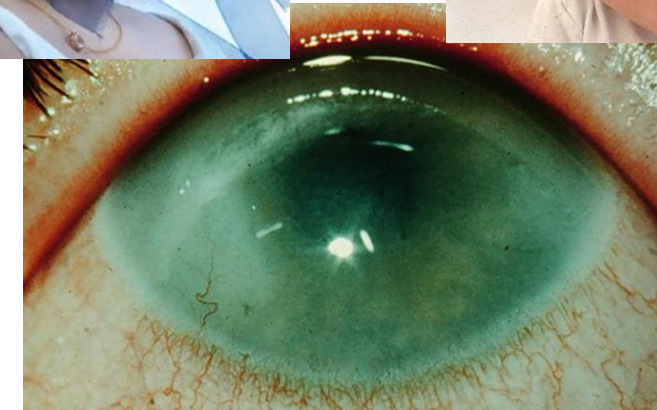
Hurler syndrome



Cleveland Clinic
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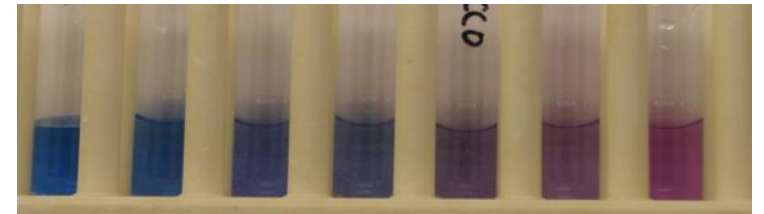
Coarse facial features, short stature, developmental delay, hepatosplenomegaly, restricted joint mobility (joint contractures)



Corneal clouding

Hurler syndrome

- Enzyme deficient: **Iduronidase**
- Substrates accumulating: **Dermatan sulfate and heparan sulfate**
- Urine is positive for glycosaminoglycans
- Fibroblast assay for the deficient enzyme is diagnostic
- Enzyme replacement therapy with **iduronidase** has been successful in many patients
- Hurler > Hurler-Scheie > Scheie

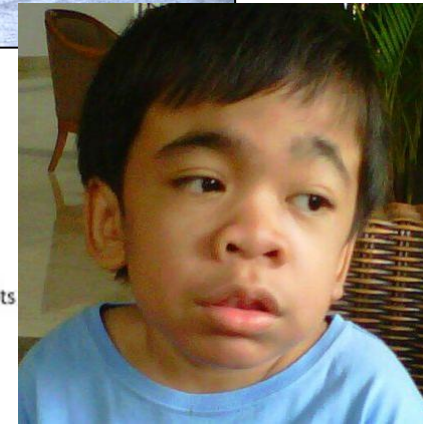
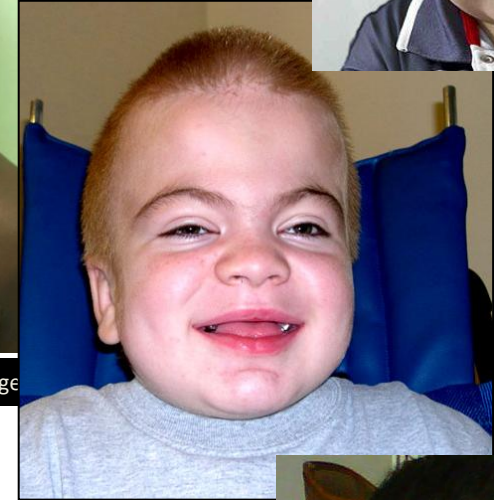




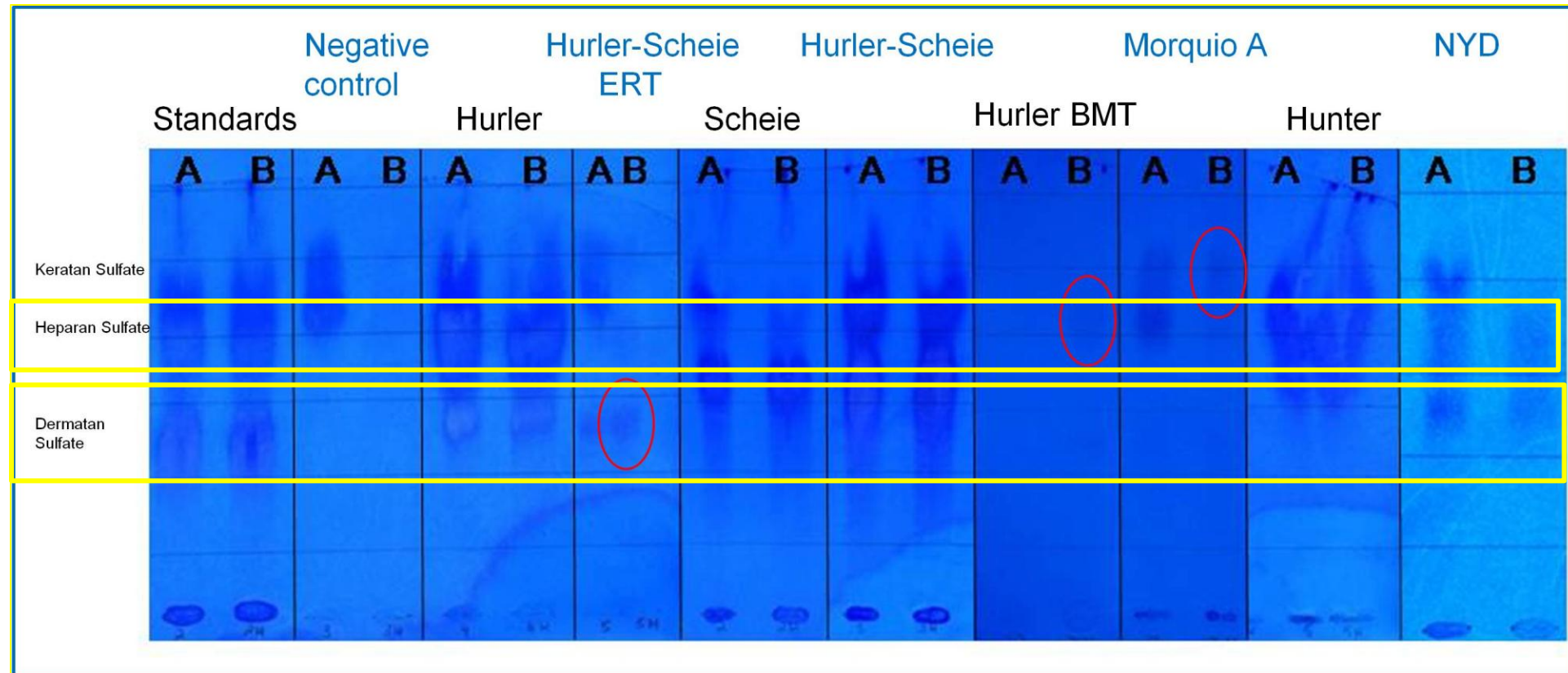
Hunter syndrome



- Milder form
- **X-linked recessive** (affects males predominantly)
- Enzyme deficient: **Iduronate sulfatase**
- Coarse facial features, hepatosplenomegaly, mild to moderate developmental delay, but **NO** corneal clouding
- Hematopoietic stem cell therapy
- Enzyme replacement study Jan 2014 suggest improvement in mobility, liver and spleen volumes reduced



Thin-layer chromatography to detect glycosaminoglycans in urine of patients with MPS

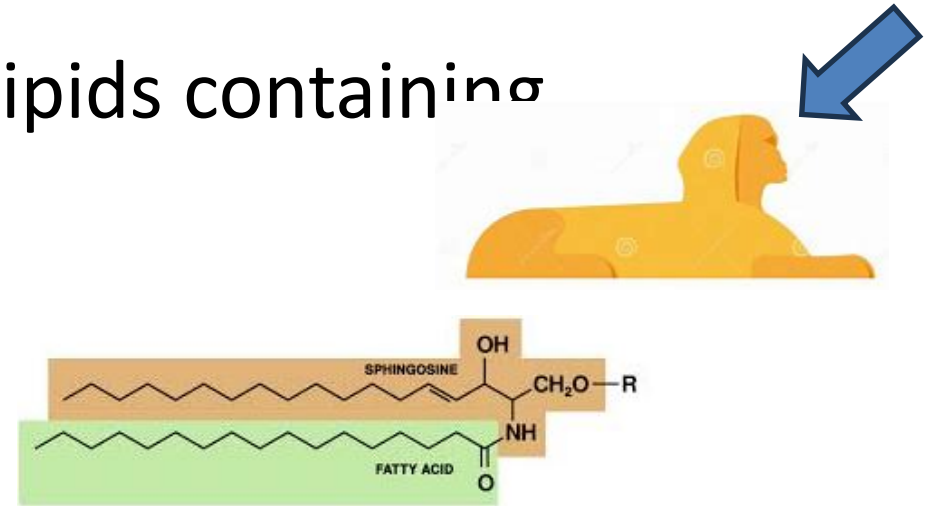


Lane A is direct sample, Lane B is sample after treatment with the enzyme Hyaluronidase (which digests hyaluronic acid and chondroitin sulfate C, resistant are heparan sulfate, dermatan sulfate, keratan sulfate & chondroitin sulphate B
ERT=enzyme replacement, BMT=bone marrow transplant, NYD= not yet diagnosed

Sphingolipidoses

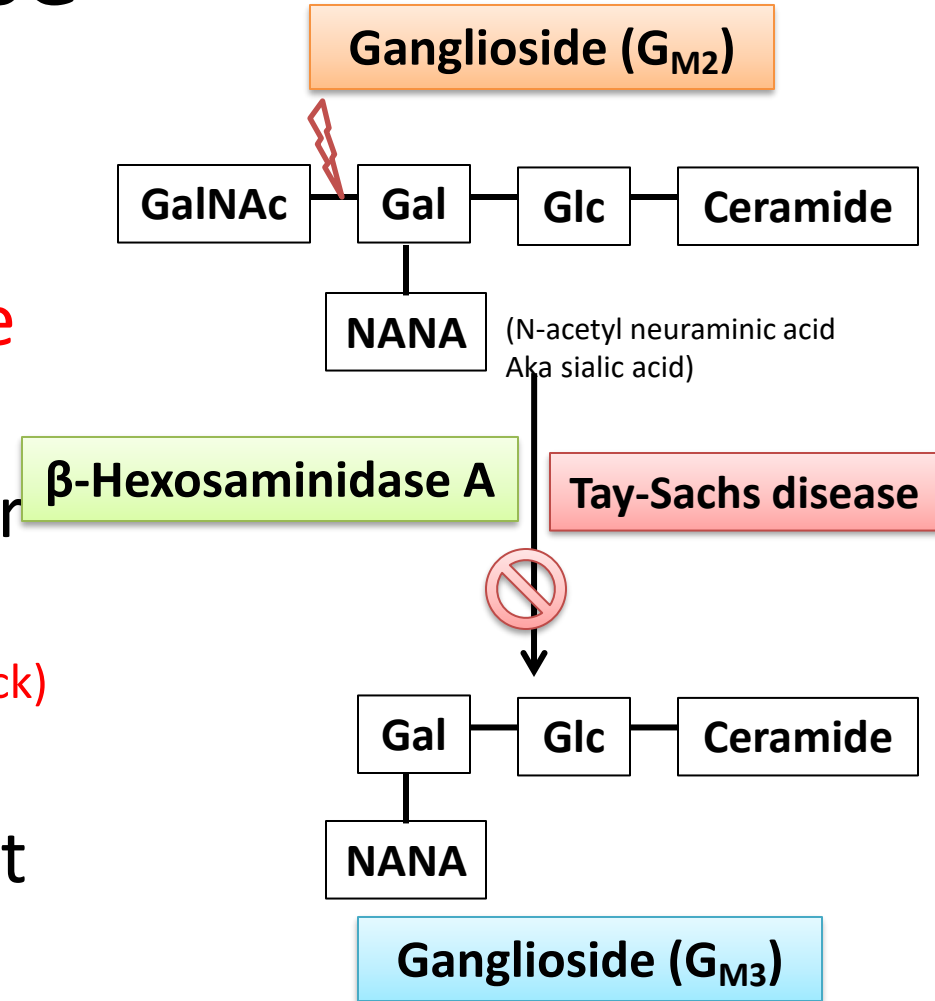
- Sphingolipids are a group of complex lipids containing sphingosine as the alcohol (not glycerol)
- Sphingosine + Fatty acid = Ceramide

- Sphingolipids are of two classes
 - **Glycosphingolipids/glycolipid** – containing a carbohydrate moiety linked to ceramide
 - **Sphingophospholipids** that contain a phosphoryl choline moiety linked to ceramide (e.g. sphingomyelin)

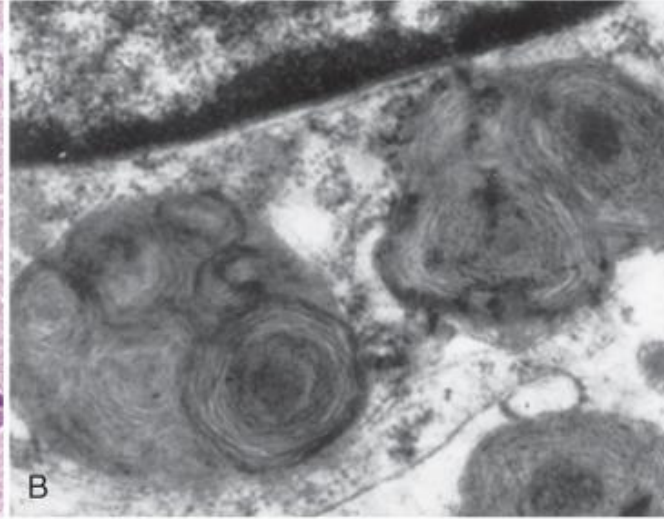
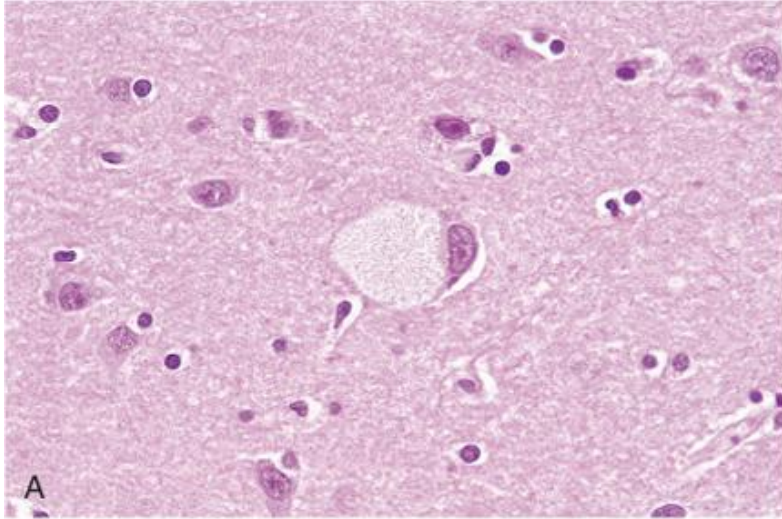


Tay-Sachs disease

- **Deficient enzyme:**
 β -Hexosaminidase A
- Accumulating substrate: **Ganglioside (G_{M2})**. Also known as gangliosidosis
- Progressive **neurodegeneration** after the age of 3-6 months, **blindness**
- **NO** hepatosplenomegaly (differentiate from Niemann-Pick)
- **Developmental milestone delay**
often regression (child not able to sit or stand in the later stages)
- Generally fatal by 2-6 years



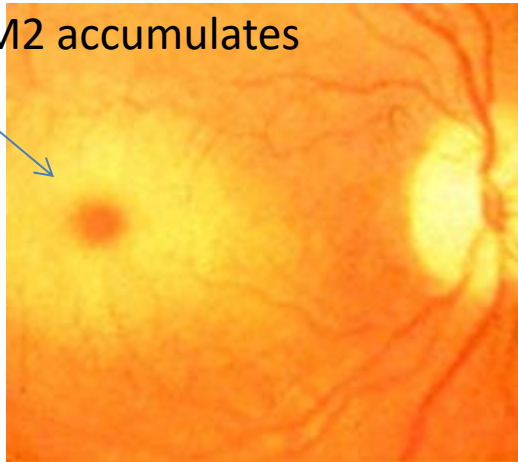
Tay-Sachs disease



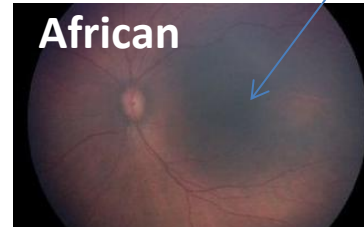
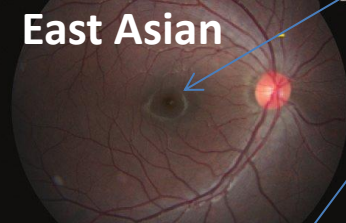
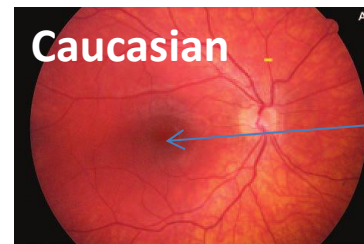
Ganglion cells in Tay-Sachs disease. **A.** Under the light microscope, a large neuron has obvious lipid vacuolation. **B,** Tay-Sachs disease. A portion of a neuron under the electron microscope shows prominent lysosomes with whorled configurations.

© Elsevier. Kumar et al: Robbins Basic Pathology 8e - www.studentconsult.com

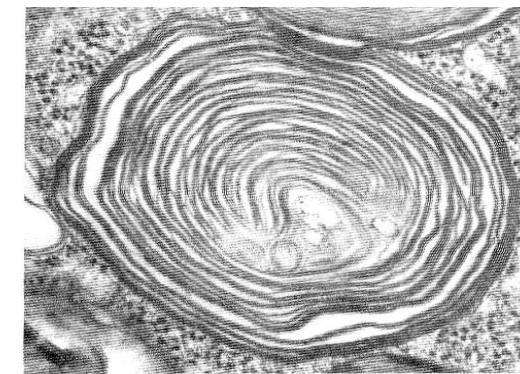
Destruction of neurons as GM2 accumulates



Cherry-red spot on macula



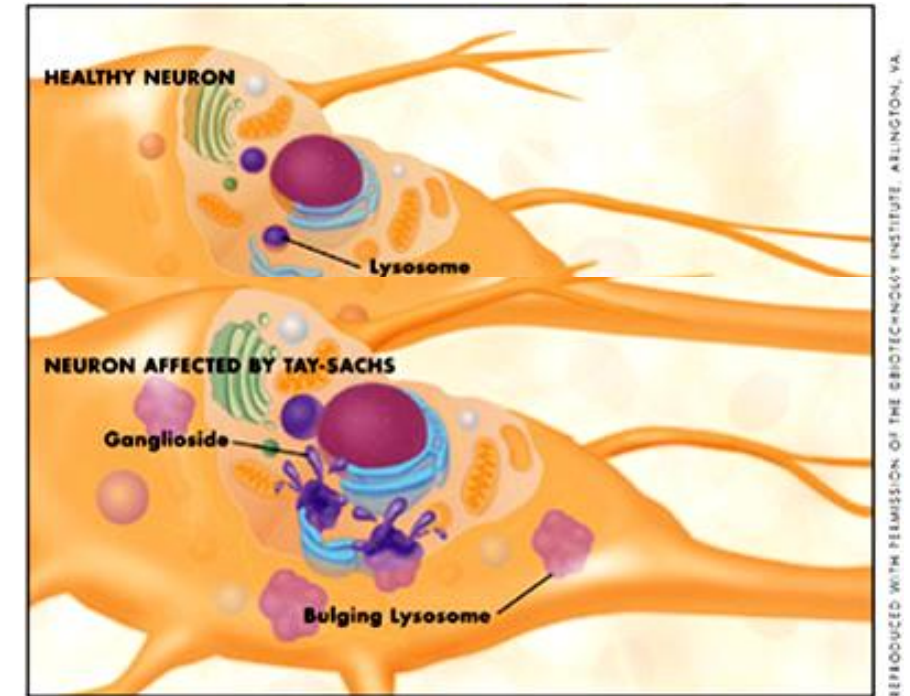
Region of highest vision



Onion-shell inclusions in lysosomes

Rochtchina et al, DOI 10.1167/iows.07-0150

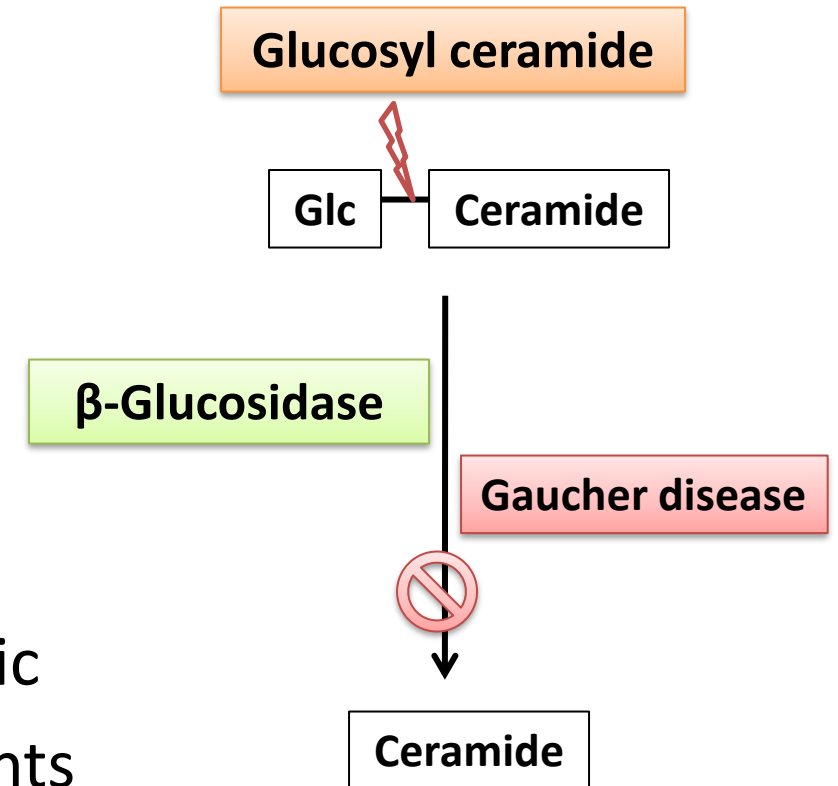
Tay-Sachs disease



- Carriers (heterozygotes) have *one copy of the mutant gene* and one copy of the normal gene
- Carriers can be detected by enzyme assays – they have about *50% hexosaminidase activity* compared to normal controls – Carriers are phenotypically normal and have no manifestations

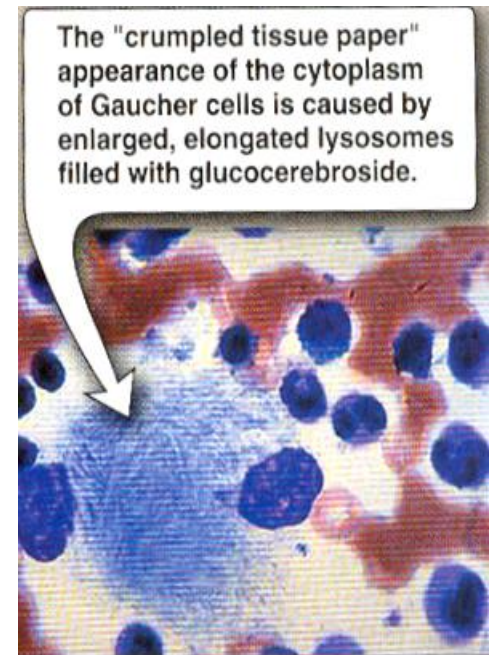
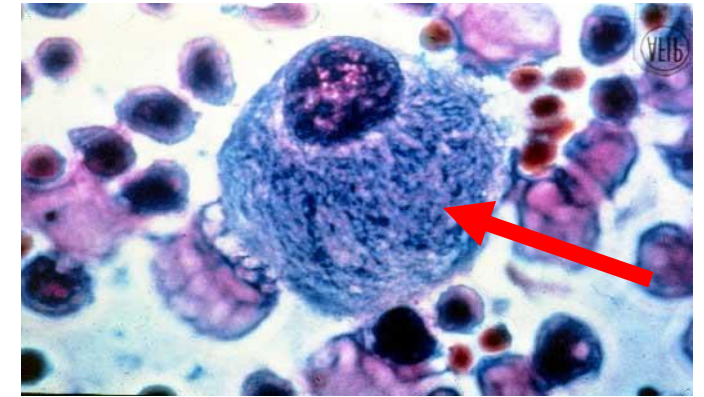
Gaucher disease

- Most common lysosomal storage disorder
- Deficient enzyme: **β -Glucosidase**
- Accumulating substrate: **Glucosyl ceramide (aka glucocerebroside)**
- Macrophages engorged with glucocerebroside
- Adult form (most common) shows no neurological damage but marked hepatosplenomegaly and osteoporosis of long bones
- “Bone Crisis” is often when adults first come to clinic
- Enzyme replacement therapy is used in many patients with Gaucher disease >



Gaucher disease

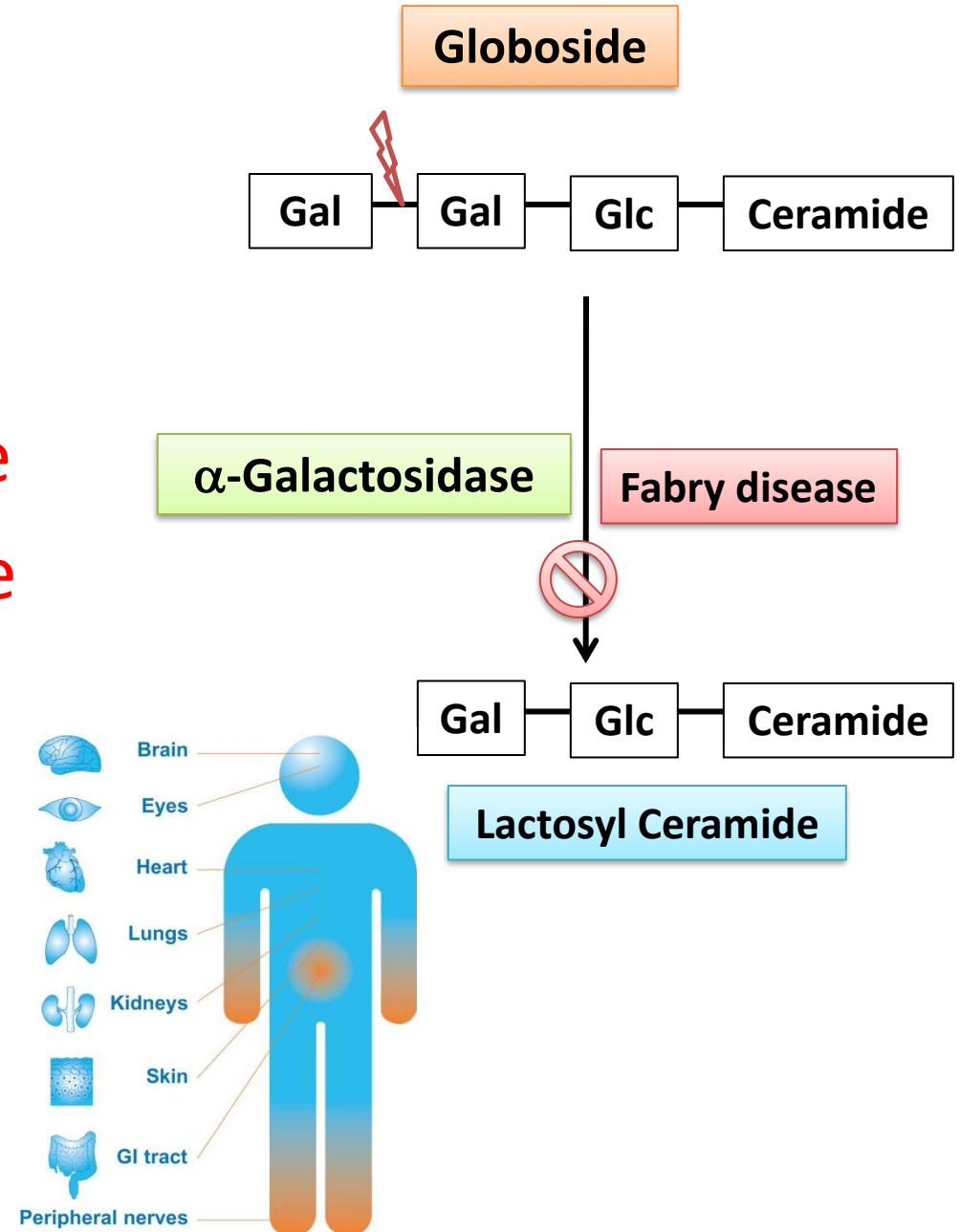
distal femur in the shape
of an *Erlenmeyer flask*



Gaucher cell

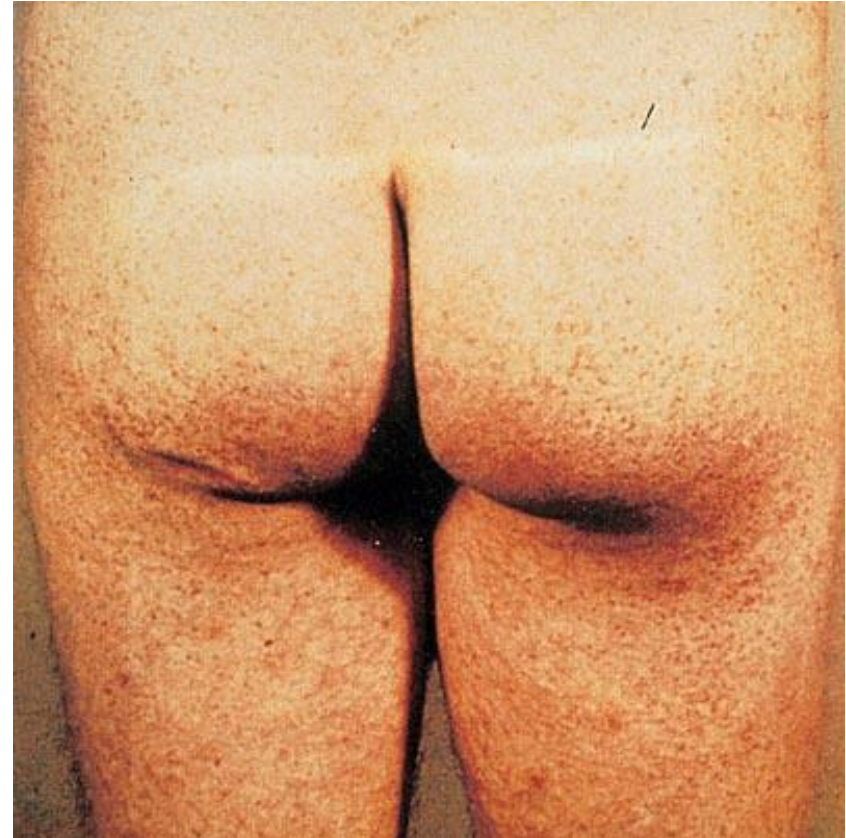
Fabry disease

- X-linked recessive disorder (Males commonly affected)
- Deficient enzyme: α -Galactosidase
- Accumulating substrate: Globoside (aka ceramide trihexoside)
- Peripheral neuropathy (tingling and burning of extremities, “acroparesthesias”)



Fabry disease

- Globoside accumulates in the blood vessels of skin, kidneys, nerves and heart
- Many children may have life threatening complications like kidney damage, heart attack and stroke
- Carrier females can show mild clinical features



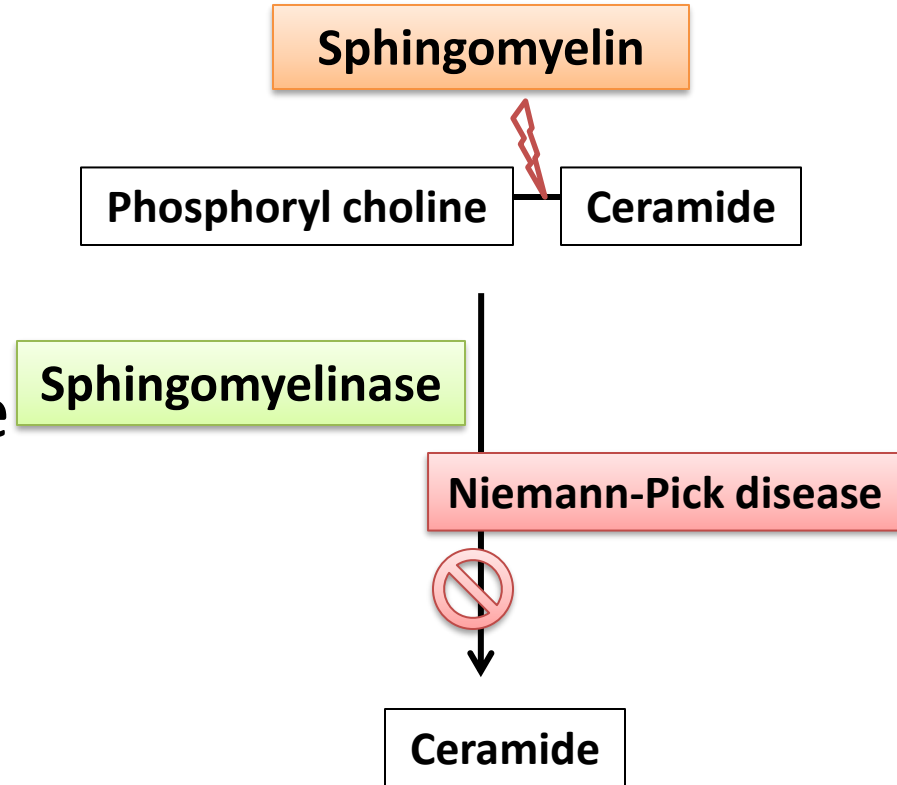
Images.MD

Manifested as skin rash 'bathing-trunk distribution'
angiokeratoma



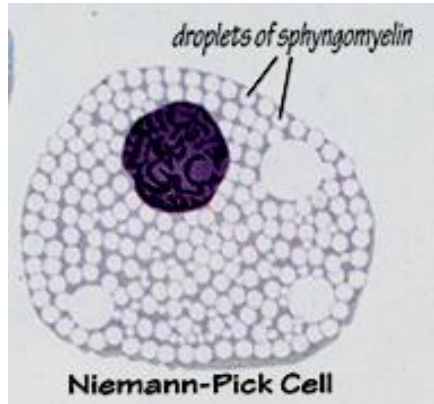
Niemann-Pick disease

- Deficient enzyme: **Sphingomyelinase**
- Accumulating substrate: **Sphingomyelin** (sphingophospholipid)
- Accumulation of sphingomyelin in the neuronal tissues
- Type A is a severe infantile form
 - Fatal by 2-3 years
 - Cherry-red spot in macula on retinal examination (blindness)
 - Hepatosplenomegaly
- Type B appears later in childhood, presents with hepatosplenomegaly



Which type is more like Tay-Sachs and which is more like Gaucher?

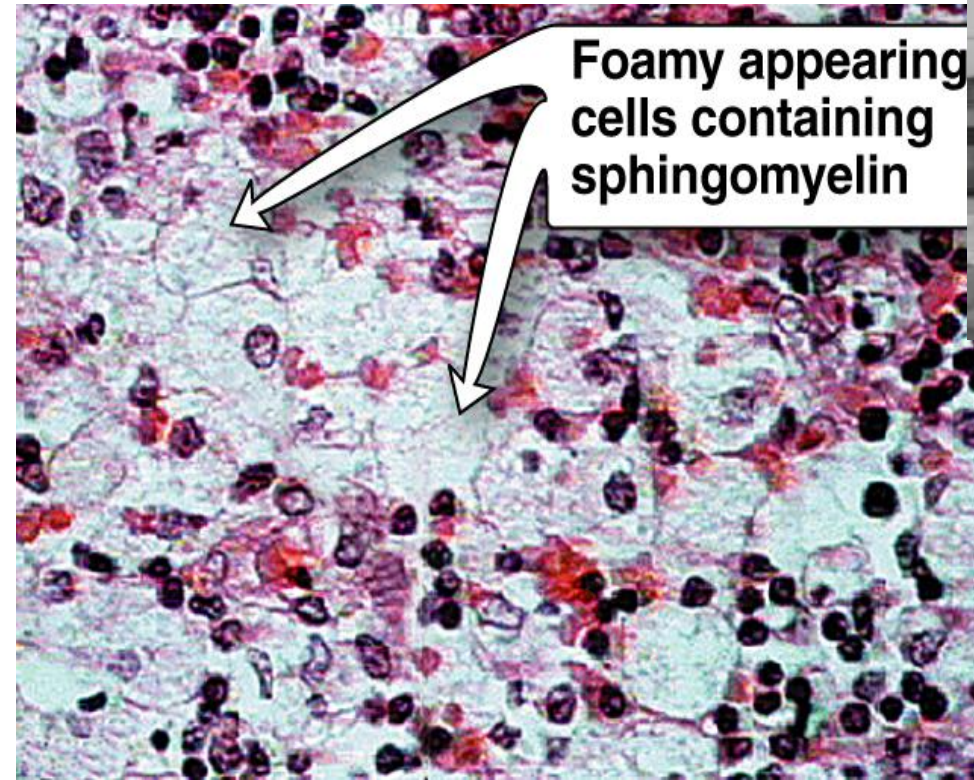
Niemann-Pick disease



Deficiency of
sphingomyelinase causes
lipid droplet accumulation-
'Foamy cell appearance'

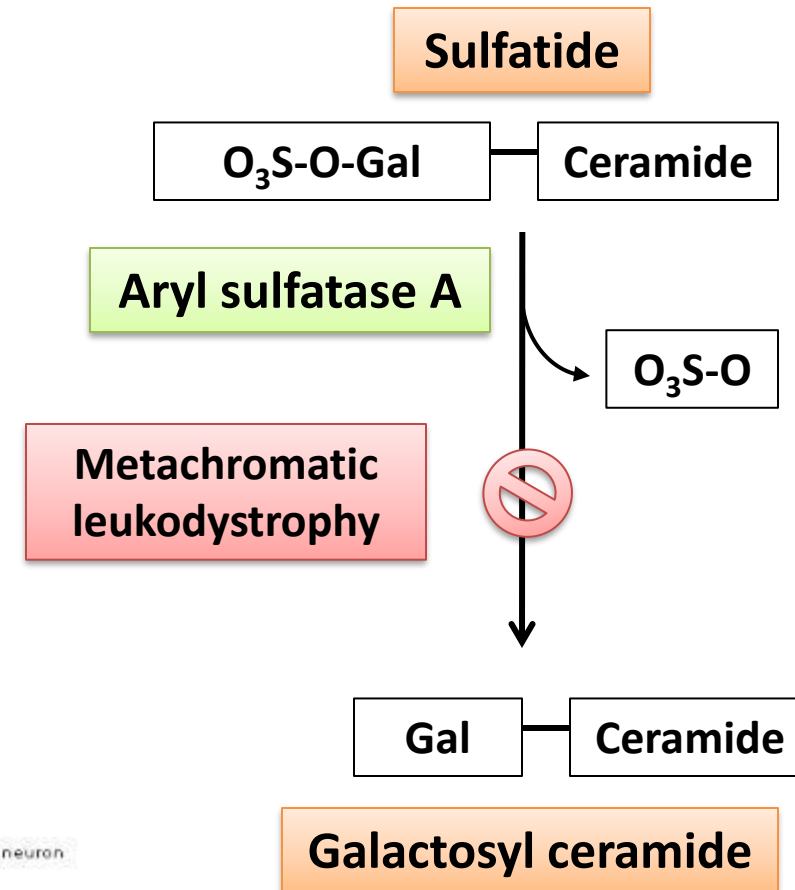
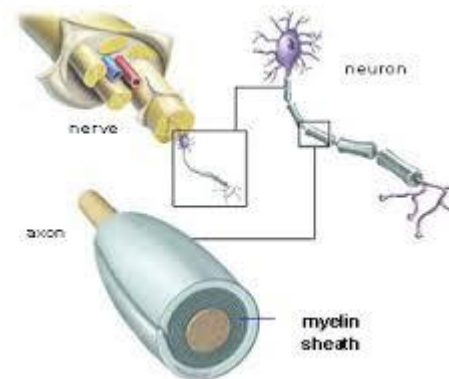


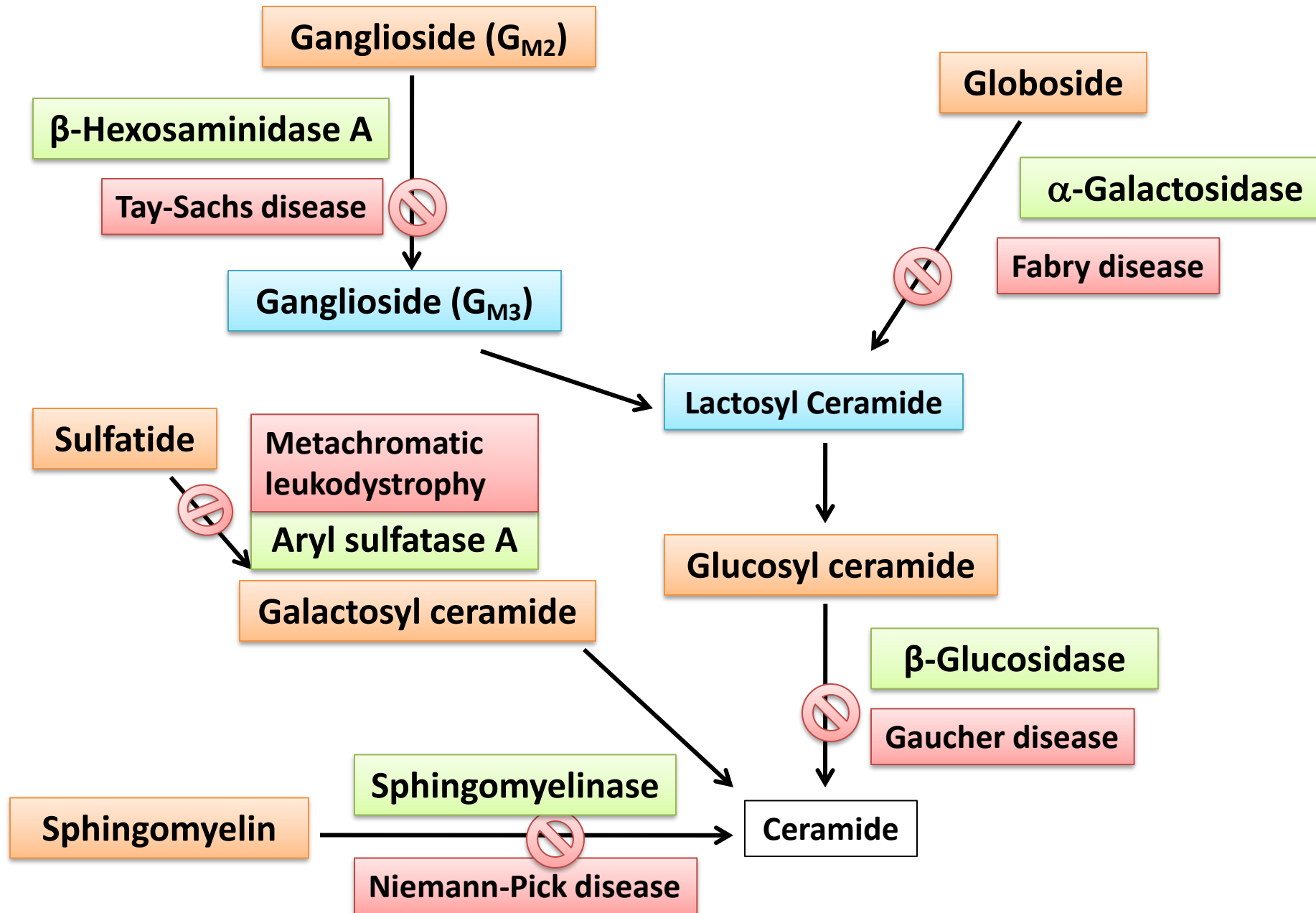
HEPATOSPLENOMEGALY



Metachromatic leukodystrophy

- Deficient enzyme: **Aryl Sulfatase A**
- Accumulating substrate: **Sulfatide** (rich in neurons)
- Progressive paralysis and demyelination





Niemann-Pick is Only disease discussed today characterized by accumulation of sphingophospholipid

Pompe disease

- A small amount of cellular glycogen is degraded by the **lysosomal acid maltase (1→4 glucosidase)**
- Generalized accumulation of **glycogen** in **heart, muscle, kidney and liver** as vacuoles in the lysosomes (glycogen storage disorder type II)
- Enzyme replacement therapy has been successful
- Glycogen accumulation in the
 - Cardiac muscle can result in cardiomegaly and death due to cardiac failure
 - Skeletal muscle can result in myopathy; hypotonia
 - Liver can result in hepatocellular damage

Pompe disease

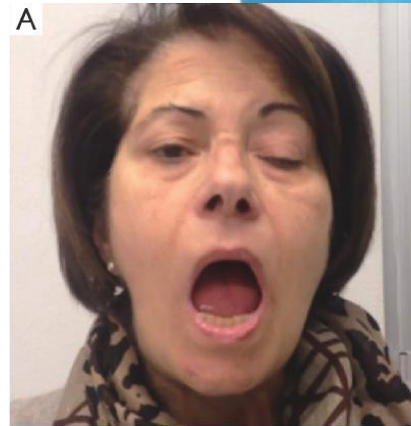
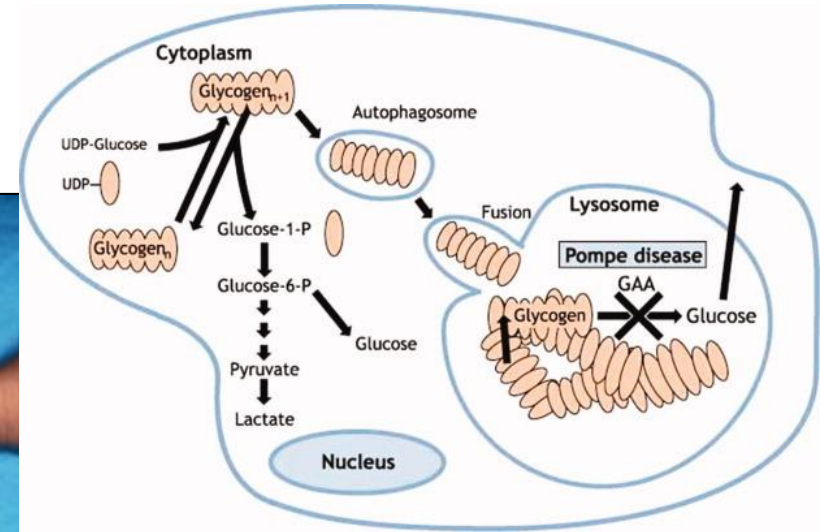
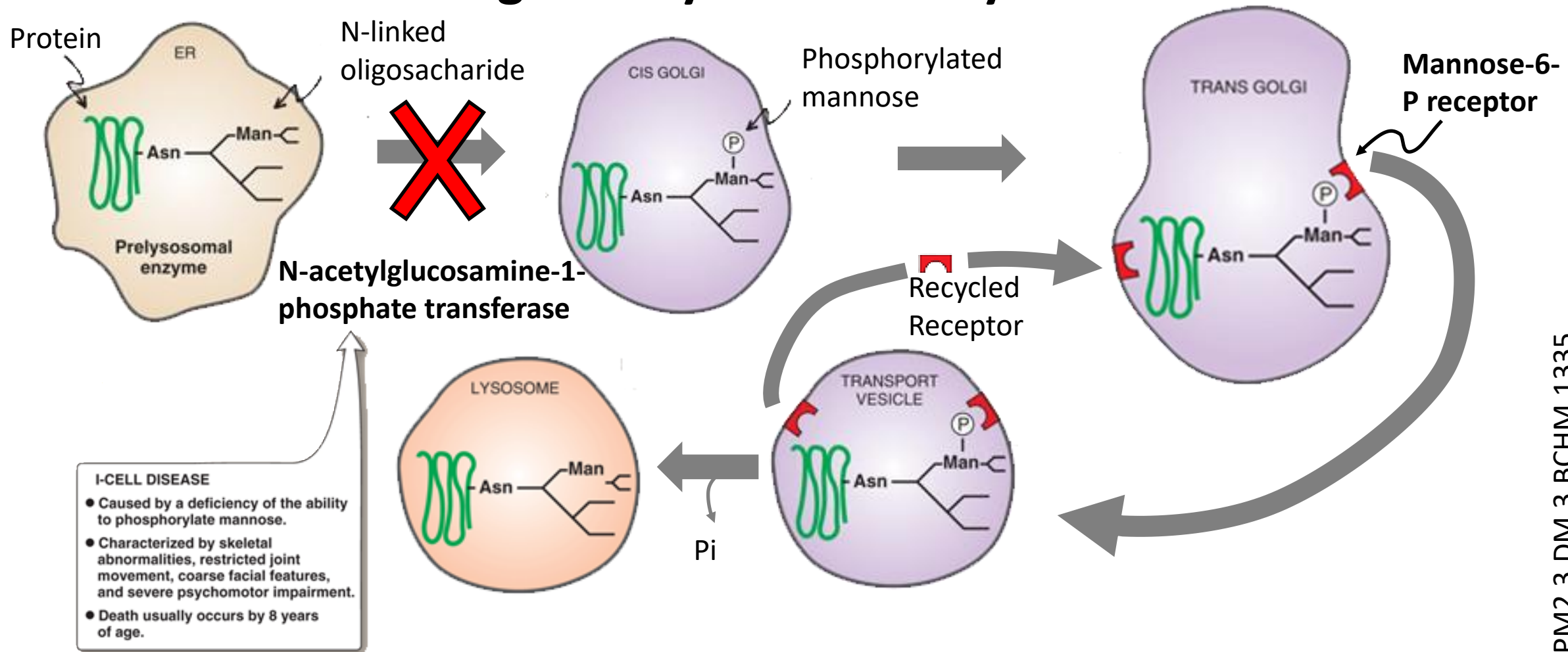


Figure 2 LOPD clinical aspects. (A) Unilateral eyelid ptosis in a 64-year-old LOPD female. (B) scapular winging in a 55-year-old LOPD

Trafficking of enzymes to the lysosomes



- Enzymes synthesized in the ER are transported to the Golgi apparatus
- In the Golgi apparatus, there is phosphorylation of mannose to form mannose-6-phosphate
- Enzymes that have a mannose-6-P are transported to the lysosomes
- Enzymes/proteins without the mannose-6-P are secreted by the cell

I-cell disease (Inclusion-cell disease)

- Children with the disease have a **defect in trafficking enzymes to the lysosomes**
- The Golgi fails to add the mannose-6-P marker to the enzymes destined for the lysosomes
- These enzymes (**without the mannose-6-P marker**) are secreted into the blood stream. An associated finding in these patients is the presence of **high** concentration of **lysosomal enzymes in the blood**
- There is a deficiency of many lysosomal enzymes, and there is accumulation of **glycosaminoglycans and sphingolipids in the lysosomes**
- A unique feature of this disease is the presence of intracytoplasmic inclusions in the fibroblasts of patients. These cells are termed '**inclusion cells**', or '**I-cells**'

I-cell disease (Inclusion-cell disease)

Clinical picture and manifestations similar to Hurler syndrome, but more severe and an earlier age of onset

Profile view of 3-year-old with I-cell disease. Growth ceased more than one year earlier. Note small orbits, proptotic eyes, full and prominent mouth caused by gingival hypertrophy, short and broad hands, stiffening of small hand joints, prominent abdomen (hepatosplenomegaly) with umbilical hernia, and limited extension of the hips and knees.

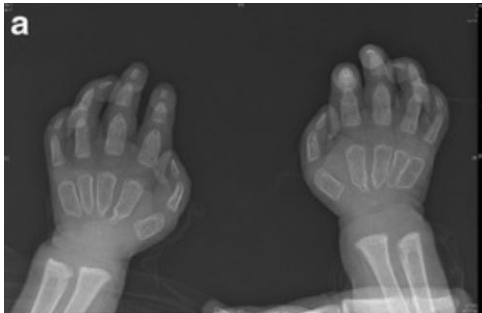
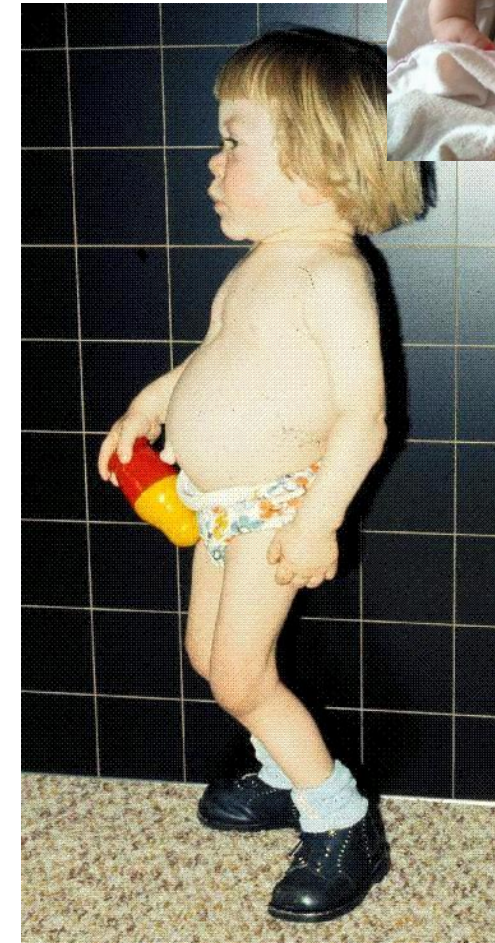
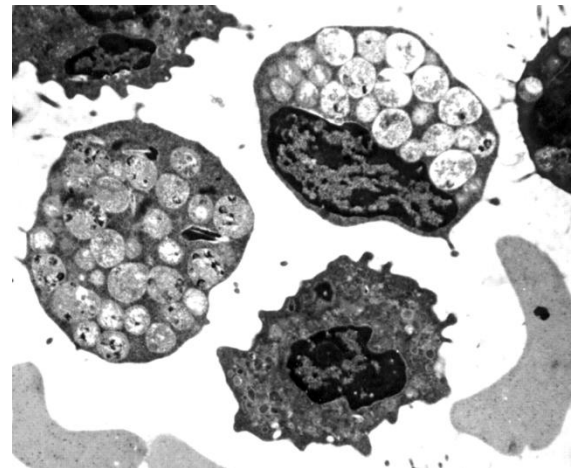
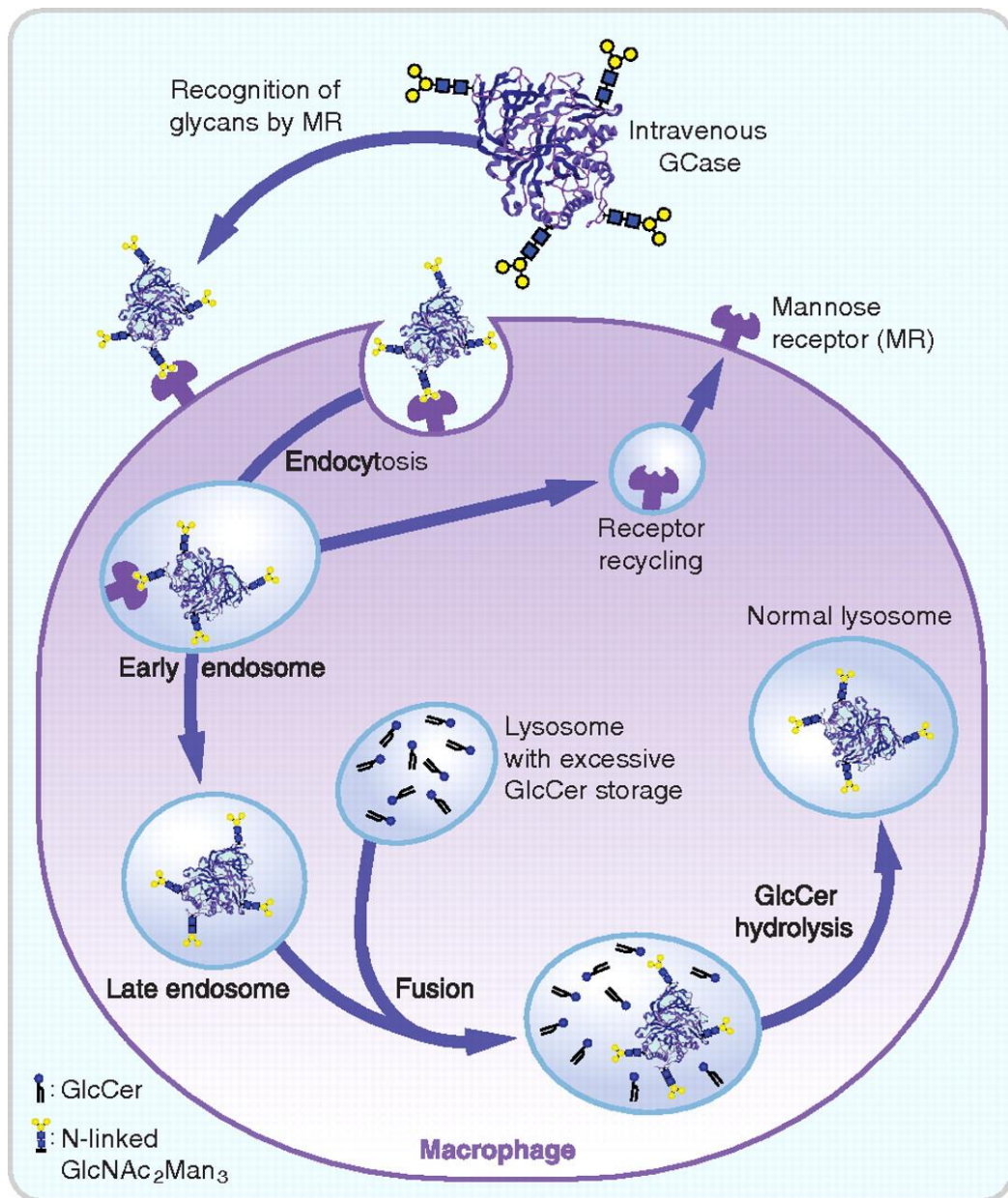


Figure 1: Clinical photograph of a two-year old girl with coarse facial features, gum hypertrophy and thick alveolar ridges



Lymphocyte appearance in a patient with I-cell disease

Enzyme Replacement Therapy



Mnemonic from the internet, all are correct

1. They're all autosomal recessive, with the exception of Fabry's and Hunter's, which are both X-linked recessive.
2. Hurler and Hunter's both start with an H, and both accumulate Heparan sulfate. (Hopefully that will help you remember that they accumulate Dermatan sulfate too).
3. Ashkenazi Jews have a higher risk of Gaucher's, Niemann-Pick, and Tay-Sachs.
4. Gaucher's spares the Brain., has crumpled tissue paper appearance.
6. "Cherry-Red" (spot on macula) is hyphenated. So are "Niemann-Pick" and "Tay-Sachs." :)
7. Hunters is X-linked. I view a Hunter shooting his bow at a X (target).
8. Niemann-Picks - Sphingomyelinase. No man picks his nose with his Sphinger (finger).
9. As above regarding Hunter's vs Hurlers. Hurlers you get corneal clouding, so you cannot be a good hunter.
10. HurLer's is alpha-L-iduronidase. Hunters, which is a similar disease, is the other one (iduronate sulfatase). "I just remember one, then know that the other one is the other one."
11. Metachromatic Leukodystrophy : Arysulfatase A deficiency
12. One way to remember Iduronidase is that it has two D's and its a hur-dle to pronounce

Other therapies

- **Migalastat** drug treatment for Fabry- pharmaceutical chaperone which helps the mutated protein fold correctly and recover some activity
- **Miglustat** drug treatment for Gaucher type 1, inhibits the formation of glycosphingolipids and this is called “substrate reduction”
- **Ambroxol** drug treatment for Gaucher, stabilizes misfolded glucocerebrosidase proteins (due to mutations)
- **Current Gene Therapy Trials for:**
 - Tay-Sachs, Fabry, Gaucher, Metachromatic leukodystrophy,
 - Some trials are Adeno-Associated Virus Gene therapy injected into brain or spine of patient
 - Some trials use Lentivirus to modify CD34+ hematopoietic stem cells and transplant into patient

Maroteaux-Lamy syndrome (MPS VI) video is 5 minutes long



<https://www.youtube.com/watch?v=jvreAK132n0>



**When you are
worn out and
frustrated,
remind yourself
why you are here!**

***Best wishes for the
exam!***